Tali Gankin

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EDUCATION

University of Southern California, Viterbi School of Engineering - Los Angeles, CA **Graduation: May 2028**
Bachelor of Science in Computer Engineering and Computer Science
Honors: Presidential Scholar Recipient (~2% acceptance rate), Viterbi Dean's List (2x), Viterbi CURVE Research Fellowship
Relevant Coursework: Data Structures and Object-Oriented Programming and Design, Discrete Methods in CS, Intro to Digital Circuits (Low-latency systems), Linear Algebra, Digital System Design with FPGAs, Embedded Systems, Web Development

SKILLS

Skills: C/C++, Verilog, Python, Figma, HTML, CSS, JavaScript, Unix/Linux, Git, Docker, Java, SQL familiarity, MS Office
Languages: Russian (Fluent), Spanish (Intermediate), Hebrew (Familiarity)

PROJECTS

Attendance Dashboard
Software Engineering Project
- Built a full-stack web dashboard (HTML, CSS, JavaScript) using Git/Docker, automating 60+ members' attendance tracking
- Applied database design concepts with the help of Figma and dynamic tracking to manage records, ensuring confidentiality
- Reduced manual work by ~2 hrs/week by automating record management workflows inspired by CRM-style data systems

BinBot
Technical Project Manager
- Programmed a Python-based algorithm to classify recyclables with sensor inputs, achieving ~90% accuracy across 50+ trials
- Created data visualizations (Excel, Python, exposure to Tableau) to communicate results clearly in a showcase (1st place)
- Applied ethical engineering principles to address UN SDGs, building a 40% more interactive education system on recycling

EXPERIENCE

CURVE Fellow (Research) **Los Angeles, CA**
Research Assistant *January 2025 - Present*
- Developed MATLAB/Python scripts for 10K+ data points, improving runtime efficiency by 20% and increasing scalability
- Applying advanced linear algebra techniques in Graph Signal Processing (GSP) to interpolate and analyze varying datasets
- Collaborating with faculty 4hrs/wk in a fast-paced, research-driven environment to balance multiple investigative tasks

Exemplary Student Research Program (Research) **Chicago, IL**
Research Assistant *August 2021 - June 2024*
- Measured material's potential effectiveness in improving battery capacity and, separately, enhancing carbon capture
- Tested 3+ hypotheses with 200+ trials using the Argonne National Lab's linear accelerator and advanced photon source
- Created three research posters, each presented annually at the international APS users meeting, to ~50 faculty and peers

Code Ninjas **Glenview, IL**
Computer Language Instructor *February 2023 - February 2024*
- Provided individualized help for over 40 children aged 6 to 14, simplifying coding logic & enhancing tutees' understanding
- Taught 5+ programming platforms, from Scratch, Repl.it, Unity, to Impact, nurturing debugging/problem-solving strategies

LEADERSHIP AND INVOLVEMENT

USC Theta Tau Engineering Society **Los Angeles, CA**
Scribe *May 2025 - Present*
- Managing sensitive organizational records for 40+ members and 100+ applicants, ensuring accuracy and confidentiality
- Conducting hourly communication on Slack through optimized 30% and increasing active participation by ~80%

Professional Chair and Social Chair *September 2024 - May 2025*
- Fostered a collaborative community of 70+ members through weekly professional workshops and networking opportunities
- Optimized allocation of $6,000+ in organizational funds to strategically boost events and strengthen member engagement

Joint Education Program - Peace Project **Los Angeles, CA**
Volunteer and Team Lead *September 2025 - Present*
- Leading event-planning workflows using collaborative tools (e.g. Microsoft Office) to coordinate outreach and curriculum
- Volunteering weekly in South Central K-12 schools, teaching 30 students mental health awareness and emotional regulation
- Collaborating with peers from 4+ majors to refine course plans, facilitating retrospectives, and increasing impact by ~25%